

LLD Case Study 11: Automated Vending Machine

Target Patterns: State Pattern, State Machine Transitions

Key Engineering Focus: Hardware lifecycle modeling, coin balance tracking, inventory deduction, and change calculation.

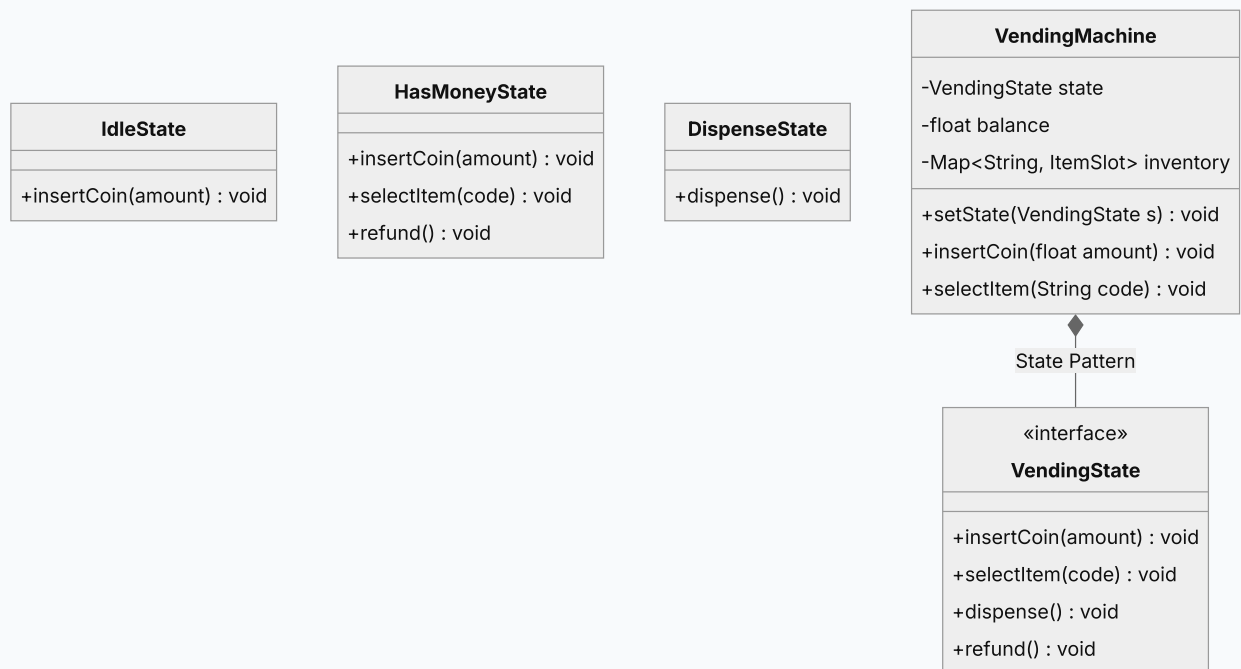
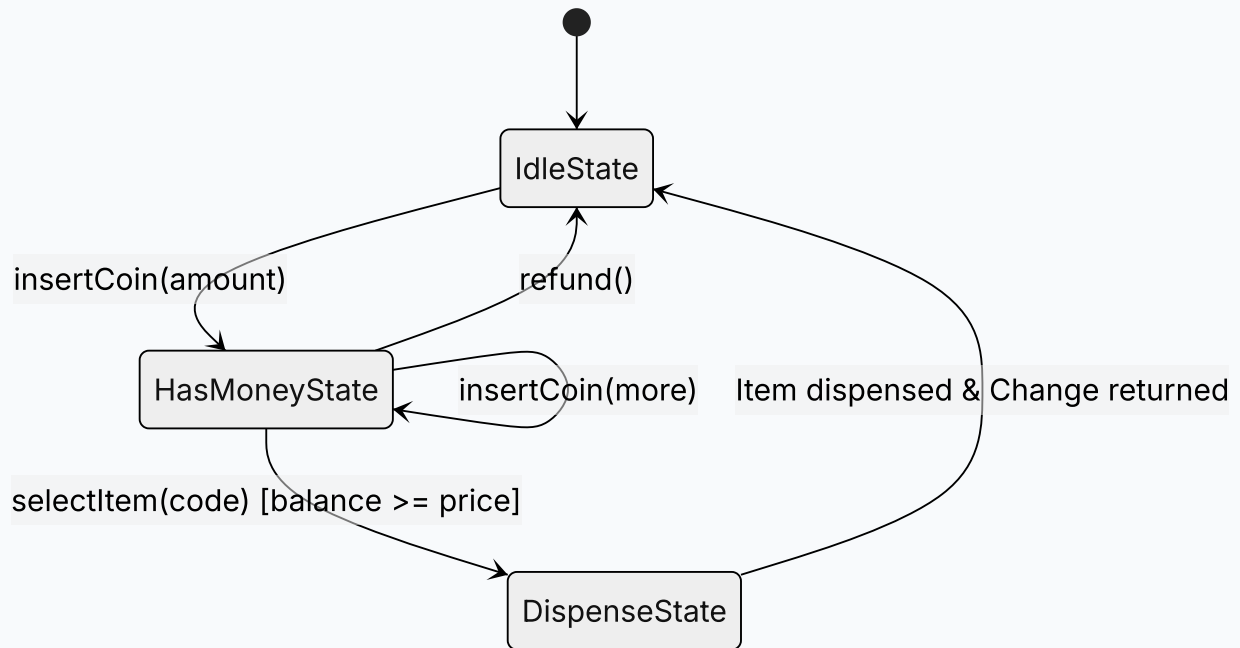
1. Problem Statement & Functional Requirements

Design an automated snack and beverage vending machine.

Requirements:

1. **Item Catalog & Inventory:** Slots storing items with fixed prices and quantities.
 2. **State Pattern:** The machine transitions through strict states:
 - **IDLE** : Awaiting money insertion.
 - **HAS_MONEY** : Tracking user balance; accepts further coins or item selection.
 - **DISPENSING** : Deducting inventory and dispensing item.
 - **SOLD_OUT** : Out of stock state.
 3. **Change Return:** Computes exact change if user inserts more than the item price.
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2. Architecture & State Diagram



3. Production-Grade Python Implementation

